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(1)
Module 2
Vector Integral Theorem

Green's Theorem

Let R be a simply connected plane region whose boundary is a simple closed, piecewise smooth curve C oriented anti-clockwise. If $f(x, y)$ and $g(x, y)$ are continuous and have continuous first order partial derivatives on some open set containing R , then

$$\oint_C f(x, y) dx + g(x, y) dy = \iint_R \left(\frac{\partial g}{\partial x} - \frac{\partial f}{\partial y} \right) dA$$

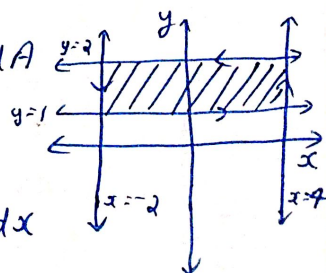
- 1) Use Green's theorem to evaluate $\oint_C 4xy dx + 2xy dy$ where C is the rectangle bounded by $x = -2$, $x = 4$, $y = 1$ and $y = 2$

Soln :- Here $f(x, y) = 4xy$ and $g(x, y) = 2xy$

$$\Rightarrow \frac{\partial f}{\partial y} = 4x \qquad \Rightarrow \frac{\partial g}{\partial x} = 2y$$

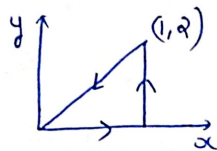
By Green's theorem,

$$\begin{aligned} \oint_C 4xy dx + 2xy dy &= \iint_R (2y - 4x) dA \\ &= \int_{x=-2}^4 \int_{y=1}^2 (2y - 4x) dy dx \end{aligned}$$



$$\begin{aligned}
 &= \int_{x=-2}^4 \left[y^2 - 4xy \right]_1^2 dx = \int_{x=-2}^4 (3 - 4x) dx \\
 &= \left[3x - 2x^2 \right]_{-2}^4 \\
 &= \underline{\underline{-6}}
 \end{aligned}$$

2) Use Green's theorem to evaluate $\oint_C x^2 y dx + x dy$ along the triangular path given by



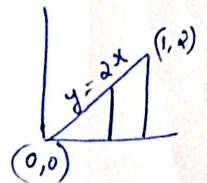
Soln : Here $f(x,y) = x^2 y$ and $g(x,y) = x$

$$\Rightarrow \frac{\partial f}{\partial y} = x^2 \quad \frac{\partial g}{\partial x} = 1$$

$$\oint_C x^2 y dx + x dy = \iint_R (1 - x^2) dA$$

$$= \int_{x=0}^1 \int_{y=0}^{2x} (1 - x^2) dy dx$$

$$= \int_0^1 \left[(1 - x^2) y \right]_0^{2x} dx = \int_0^1 (2x - 2x^3) dx = \underline{\underline{\frac{1}{2}}}$$



3) Using Green's theorem, find the work done by the force field $\vec{F}(x,y) = (e^x - y^3)\hat{i} + (\cos y + x^3)\hat{j}$ on a particle that travels once around the unit circle $x^2 + y^2 = 1$ in the counter clockwise direction.

(3)

Soln: $W = \oint_C \vec{F} \cdot d\vec{r} = \oint_C (e^x - y^3)dx + (\cos y + x^3)dy$

Here $f(x, y) = e^x - y^3$ and $g(x, y) = \cos y + x^3$

$$\Rightarrow \frac{\partial f}{\partial y} = -3y^2 \quad \frac{\partial g}{\partial x} = 3x^2$$

By Green's theorem, $W = \iint_R (3x^2 + 3y^2) dA$

$$= 3 \int_0^{2\pi} \int_0^1 (r^2) r dr d\theta$$

 $\theta=0 \quad r=0$

$$= 3 \int_0^{2\pi} \left[\frac{r^4}{4} \right]_0^1 d\theta = \frac{3}{4} [0]_0^{2\pi} = \underline{\underline{\frac{3\pi}{2}}}$$

- 4) Using Green's theorem evaluate $\oint_C (e^x + y^2)dx + (e^y + x^2)dy$ where C is the boundary of the region between $y = x^2$ and $y = 2x$.

Soln: $f(x, y) = e^x + y^2$ and $g(x, y) = e^y + x^2$

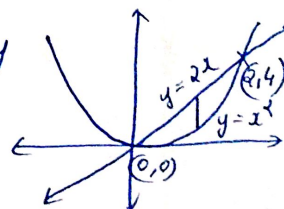
$$\Rightarrow \frac{\partial f}{\partial y} = 2y \quad \frac{\partial g}{\partial x} = 2x$$

By Green's theorem, $\oint_C (e^x + y^2)dx + (e^y + x^2)dy$

$$= \iint_R (2x - 2y) dA$$

$$= 2 \int_{x=0}^2 \int_{y=x^2}^{2x} (x - y) dy dx = 2 \int_0^2 \left[xy - \frac{y^2}{2} \right]_{x^2}^{2x} dx$$

$$= 2 \int_0^2 \left(-x^3 + \frac{x^4}{2} \right) dx = \underline{\underline{\frac{-8}{5}}}$$



(4)

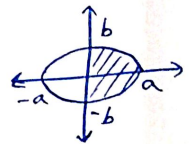
To find area using Green's Theorem

$$\text{Area of the region, } R = \iint_R dA$$

$$= \oint_C x dy \quad \text{or} \quad \oint_C (-y) dx$$

- 1) Use line integral to find the area enclosed by the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

Soln : Area = $\oint_C x dy = 2 \int_{-b}^b \frac{a}{b} \sqrt{b^2 - y^2} dy$



$$= \frac{2a}{b} \int_{-\pi/2}^{\pi/2} \sqrt{b^2 - b^2 \sin^2 \theta} b \cos \theta d\theta$$

Put $y = b \sin \theta$
 $dy = b \cos \theta d\theta$

$$= \frac{2a}{b} \int_{-\pi/2}^{\pi/2} b^2 \cos^2 \theta d\theta$$

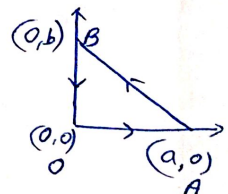
$$= \frac{2a}{b} \int_{-\pi/2}^{\pi/2} b^2 \left(\frac{1 + \cos 2\theta}{2} \right) d\theta = ab \left[\theta + \frac{\sin 2\theta}{2} \right]_{-\pi/2}^{\pi/2}$$

$$= \underline{\underline{\pi ab}}$$

- 2) Use line integral, to find the area of the triangle with vertices $(0,0)$, $(a,0)$ and $(0,b)$ where $a > 0$, $b > 0$.

Soln : Area = $\oint_C x dy = \int_{OA} x dy + \int_{AB} x dy + \int_{BO} x dy$

$\begin{matrix} OA \downarrow & AB \downarrow & BO \downarrow \\ y=0 & x=a-\frac{a}{b}y & x=0 \\ dy=0 & & dx=0 \end{matrix}$



$$= 0 + \int_{y=0}^b \left(a - \frac{a}{b}y \right) dy + 0$$

Eqn of AB
 $\frac{x-a}{0-a} = \frac{y-0}{b-0}$
 $\Rightarrow x = a - \frac{a}{b}y$

$$= \left[ay - \frac{a}{b} \frac{y^2}{2} \right]_0^b = \underline{\underline{\frac{ab}{2}}}$$

H.W Ex 15.4, Question Nos: 5, 7, 10, 11, 12, 13, 29

Exercise 15.4

- 5) Using Green's theorem, evaluate $\oint_C x \cos y \, dx - y \sin x \, dy$, where C is the square with vertices $(0,0)$, $(\frac{\pi}{2}, 0)$, $(\frac{\pi}{2}, \frac{\pi}{2})$ and $(0, \frac{\pi}{2})$.
- 7) Using Green's theorem, evaluate $\oint_C (x^2 - y) \, dx + x \, dy$, where C is the circle $x^2 + y^2 = 4$.
- 10) Using Green's theorem, evaluate $\oint_C x^2 y \, dx - y^2 x \, dy$, where C is the boundary of the region in the first quadrant, enclosed between the coordinate axes and the circle $x^2 + y^2 = 16$.
- 11) Using Green's theorem, evaluate $\oint_C \tan^{-1} y \, dx - \frac{y^2 x}{1+y^2} \, dy$, where C is the square with vertices $(0,0)$, $(1,0)$, $(1,1)$ and $(0,1)$.
- 12) Using Green's theorem, evaluate $\oint_C \cos x \sin y \, dx + \sin x \cos y \, dy$, where C is the triangle with vertices $(0,0)$, $(3,3)$ and $(0,3)$.
- 13) Using Green's theorem, evaluate $\oint_C x^2 y \, dx + (y + x y^2) \, dy$, where C is the boundary of the region enclosed by $y = x^2$ and $x = y^2$.

29) Use Green's theorem to find the work done by the force field

$\vec{F}(x,y) = xy\hat{i} + \left(\frac{1}{2}x^2 + xy\right)\hat{j}$ on a particle that starts at $(5,0)$, traverses the upper semi circle $x^2 + y^2 = 25$ and returns to its starting point along the x -axis

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(5)

Surface Integral

Let $f(x, y, z)$ be a continuous function defined on a smooth surface σ . Divide σ into n subsections $\sigma_1, \sigma_2, \dots, \sigma_n$ with areas $\Delta S_1, \Delta S_2, \dots, \Delta S_n$ respectively.

Let (x_i, y_i, z_i) be any point on the i^{th} section.

Then the surface integral of $f(x, y, z)$ over σ is

$$\iint_{\sigma} f(x, y, z) dS = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i, y_i, z_i) \Delta S_i$$

where $\Delta S_i \rightarrow 0$

To evaluate surface integral

Let σ be a surface with equation $z = g(x, y)$ and the region R be its projection on the xy -plane. Then

$$\iint_{\sigma} f(x, y, z) dS = \iint_R f(x, y, g(x, y)) \sqrt{\left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2 + 1} dA$$

For $y = g(x, z)$ and R its projection on the xz -plane

$$\iint_{\sigma} f(x, y, z) dS = \iint_R f(x, g(x, z), z) \sqrt{\left(\frac{\partial y}{\partial x}\right)^2 + \left(\frac{\partial y}{\partial z}\right)^2 + 1} dA$$

For $x = g(y, z)$ and R its projection on the yz -plane

$$\iint_{\sigma} f(x, y, z) dS = \iint_R f(g(y, z), y, z) \sqrt{\left(\frac{\partial x}{\partial y}\right)^2 + \left(\frac{\partial x}{\partial z}\right)^2 + 1} dA$$

(6)

- 1) Evaluate the surface integral $\iint_{\sigma} xz \, dS$ where σ is the part of the plane $x+y+z=1$ that lies in the first octant.

Soln:- $\sigma: z = 1-x-y$

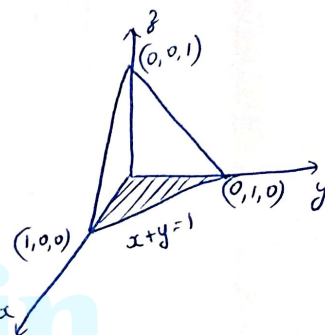
$$\iint_{\sigma} xz \, dS = \iint_R x(1-x-y) \sqrt{(-1)^2 + (-1)^2 + 1} \, dA$$

$$\begin{cases} z = 1-x-y \\ \frac{\partial z}{\partial x} = -1 \\ \frac{\partial z}{\partial y} = -1 \end{cases}$$

$$= \sqrt{3} \int_{x=0}^1 \int_{y=0}^{(1-x)} (x-x^2-xy) \, dy \, dx$$

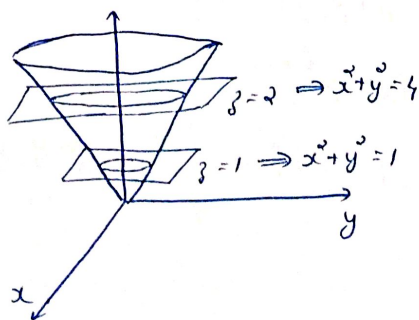
$$= \sqrt{3} \int_0^1 \left[xy - x^2y - \frac{xy^2}{2} \right]_0^{1-x} dx$$

$$= \sqrt{3} \int_0^1 \left(\frac{x}{2} - x^2 + \frac{x^3}{2} \right) dx = \sqrt{3} \left[\frac{x^2}{4} - \frac{x^3}{3} + \frac{x^4}{8} \right]_0^1 = \frac{\sqrt{3}}{24}$$

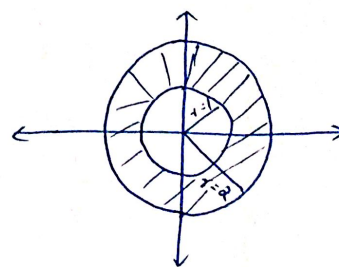


- 2) Evaluate the surface integral $\iint_{\sigma} y^2 z^2 \, dS$ where σ is the part of the cone $z = \sqrt{x^2+y^2}$ that lies between the planes $z=1$ and $z=2$.

Soln:-



Projection R



(7)

$$\sigma: z = \sqrt{x^2 + y^2} \implies \frac{\partial z}{\partial x} = \frac{x}{\sqrt{x^2 + y^2}} \quad \text{and} \quad \frac{\partial z}{\partial y} = \frac{y}{\sqrt{x^2 + y^2}}$$

$$\therefore \iint_{\sigma} y^2 z^2 \, dS = \iint_R y^2 (x^2 + y^2) \sqrt{\frac{x^2}{x^2 + y^2} + \frac{y^2}{x^2 + y^2} + 1} \, dA$$

$$= \sqrt{2} \iint_R y^2 (x^2 + y^2) \, dA$$

$$= \sqrt{2} \int_{\theta=0}^{2\pi} \int_{r=1}^2 (r \sin \theta)^2 (r^2) r \, dr \, d\theta$$

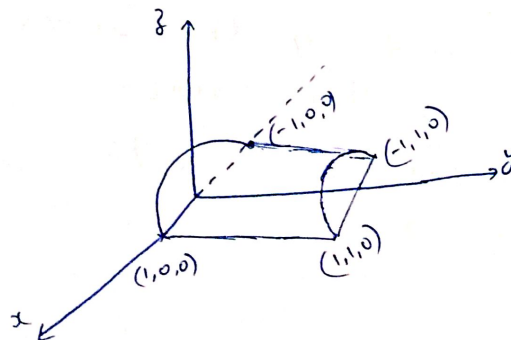
$$= \sqrt{2} \int_0^{2\pi} (\sin^2 \theta) \left(\frac{r^6}{6} \right) \Big|_1^2 \, d\theta$$

$$= \sqrt{2} \left(\frac{63}{6} \right) \int_0^{2\pi} \left(\frac{1 - \cos 2\theta}{2} \right) d\theta$$

$$= \frac{\sqrt{2}}{2} \left(\frac{63}{6} \right) \left(\theta - \frac{\sin 2\theta}{2} \right) \Big|_0^{2\pi} = \underline{\underline{\frac{21\pi}{\sqrt{2}}}}$$

3) Evaluate $\iint_{\sigma} x^2 y \, dS$ where σ is the portion of the cylinder $x^2 + z^2 = 1$ between the planes $y=0$, $y=1$ and above the xy -plane.

Soln:-



(8)

$$\sigma: x^2 + z^2 = 1 \Rightarrow z = \sqrt{1-x^2}$$

$$\frac{\partial z}{\partial x} = \frac{-x}{\sqrt{1-x^2}} \quad \text{and} \quad \frac{\partial z}{\partial y} = 0$$

$$\therefore \iint_{\sigma} x^2 y \, dS = \iint_R x^2 y \sqrt{\left(\frac{-x}{\sqrt{1-x^2}}\right)^2 + 0 + 1} \, dA$$

$$= \iint_R x^2 y \sqrt{\frac{1}{1-x^2}} \, dA$$

$$= \int_{x=-1}^1 \int_{y=0}^1 \left(\frac{x^2}{\sqrt{1-x^2}} y \right) dy \, dx$$

$$= \int_{-1}^1 \frac{x^2}{\sqrt{1-x^2}} \left(\frac{y^2}{2} \right)_0^1 dx$$

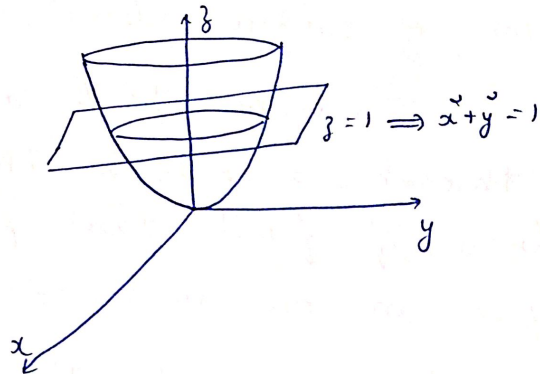
$$= \frac{1}{2} \int_{-1}^1 \frac{x^2}{\sqrt{1-x^2}} dx \quad \begin{array}{l} \text{Put } x = \sin \theta \\ dx = \cos \theta \, d\theta \end{array}$$

$$= \frac{1}{2} \int_{-\pi/2}^{\pi/2} \frac{\sin^2 \theta}{\cos \theta} \cos \theta \, d\theta$$

$$= \frac{1}{2} \int_{-\pi/2}^{\pi/2} \frac{1 - \cos 2\theta}{2} \, d\theta = \frac{1}{4} \left[\theta - \frac{\sin 2\theta}{2} \right]_{-\pi/2}^{\pi/2} = \underline{\underline{\frac{\pi}{4}}}$$

- 4) Find the mass of the lamina σ that is the portion of the paraboloid $z = x^2 + y^2$ below the plane $z = 1$ with constant mass density δ_0 .

(9)

Soln :-

$$\text{Mass} = \iint_{\sigma} \delta_0 dS = \iint_R \delta_0 \sqrt{(2x)^2 + (2y)^2 + 1} dA$$

$$= \delta_0 \iint_R \sqrt{4(x^2 + y^2) + 1} dA$$

$$= \delta_0 \int_{\theta=0}^{2\pi} \int_{r=0}^1 \sqrt{4r^2 + 1} r dr d\theta$$

$$= \delta_0 \int_0^{2\pi} \int_{t=1}^5 \left(\sqrt{t} \frac{dt}{8} \right) d\theta$$

$$= \frac{\delta_0}{8} \int_0^{2\pi} \left(\frac{2}{3} t^{3/2} \right)_1^5 d\theta$$

$$= \frac{\delta_0}{12} \int_0^{2\pi} (5^{3/2} - 1) d\theta = \frac{\delta_0}{12} (5^{3/2} - 1) [\theta]_0^{2\pi}$$

$$= \frac{\delta_0}{6} (5^{3/2} - 1) \pi$$

$$\begin{aligned} z &= x^2 + y^2 \\ \frac{\partial z}{\partial x} &= 2x \\ \frac{\partial z}{\partial y} &= 2y \end{aligned}$$

$$\begin{aligned} \text{Put } 4r^2 + 1 &= t \\ 8r dr &= dt \\ r dr &= \frac{dt}{8} \end{aligned}$$

H.W

Ex 15.5, Question Nos: 1, 2, 4, 6, 27, 29, 30, 31

Exercise 15.5

7) Evaluate the surface integral $\iint_{\sigma} f(x, y, z) \, dS$

1) $f(x, y, z) = z^2$, σ is the portion of the cone $z = \sqrt{x^2 + y^2}$ between the planes $z=1$ and $z=2$.

2) $f(x, y, z) = xy$, σ is the portion of the plane $x+y+z=1$ lying in the first octant.

4) $f(x, y, z) = (x^2 + y^2)z$, σ is the portion of the sphere $x^2 + y^2 + z^2 = 4$ above the plane $z=1$.

6) $f(x, y, z) = x+y$, σ is the portion of the plane $z = 6 - 2x - 3y$ in the first octant.

II) Find the mass of the lamina with constant density δ .

27) The lamina that is the portion of the circular cylinder $x^2 + z^2 = 4$ that lies directly above the rectangle $R = \{(x, y) : 0 \leq x \leq 1, 0 \leq y \leq 4\}$ in the xy -plane.

29) Find the mass of the lamina that is the portion of the surface $y^2 = 4-z$ between the planes $x=0$, $x=3$, $y=0$ and $y=3$ if the density is $\delta(x, y, z) = y$.

30) Find the mass of the lamina that is the portion of the cone $z = \sqrt{x^2 + y^2}$ between

$z=1$ and $z=4$ if the density is

$$s(x, y, z) = x^2 z.$$

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(10)

Application of surface integral :- Flux

Imagine that a fluid is flowing through a surface. Then flux is the volume of fluid that passes through the surface in one unit of time.

Let $F(x, y, z)$ be the vector field at a point (x, y, z) on the surface σ . Then

flux $\phi = \iint_{\sigma} \vec{F} \cdot \hat{n} \, dS$ where n is the unit normal vector towards the positive side of σ .

Let σ be the surface with equation $z = g(x, y)$ and R be its projection on the xy -plane.

$$\text{Then } \phi = \iint_{\sigma} \vec{F} \cdot \hat{n} \, dS = \iint_R \vec{F} \cdot \left(-\hat{i} \frac{\partial z}{\partial x} - \hat{j} \frac{\partial z}{\partial y} + \hat{k} \right) dA$$

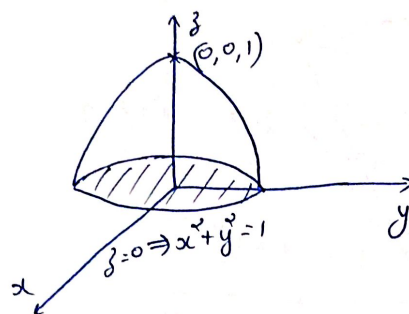
(oriented up)

$$\text{or } \iint_R \vec{F} \cdot \left(\hat{i} \frac{\partial z}{\partial x} + \hat{j} \frac{\partial z}{\partial y} - \hat{k} \right) dA$$

(oriented down)

- 1) Let σ be the portion of the surface $z = 1 - x^2 - y^2$ that lies above the xy -plane and suppose that σ is oriented up. Find the flux of the vector field, $F(x, y, z) = x\hat{i} + y\hat{j} + z\hat{k}$

(11)

Soln:

$$\sigma: z = 1 - x^2 - y^2$$

$$\frac{\partial z}{\partial x} = -2x$$

$$\frac{\partial z}{\partial y} = -2y$$

$$\phi = \iint_{\sigma} \vec{F} \cdot \hat{n} \, dS = \iint_R \vec{F} \cdot \left(-\hat{i} \frac{\partial z}{\partial x} - \hat{j} \frac{\partial z}{\partial y} + \hat{k} \right) dA$$

$$= \iint_R (x\hat{i} + y\hat{j} + z\hat{k}) \cdot (-\hat{i}(2x) - \hat{j}(2y) + \hat{k}) dA$$

$$= \iint_R (2x^2 + 2y^2 + z) dA$$

$$= \iint_R [2x^2 + 2y^2 + (1 - x^2 - y^2)] dA$$

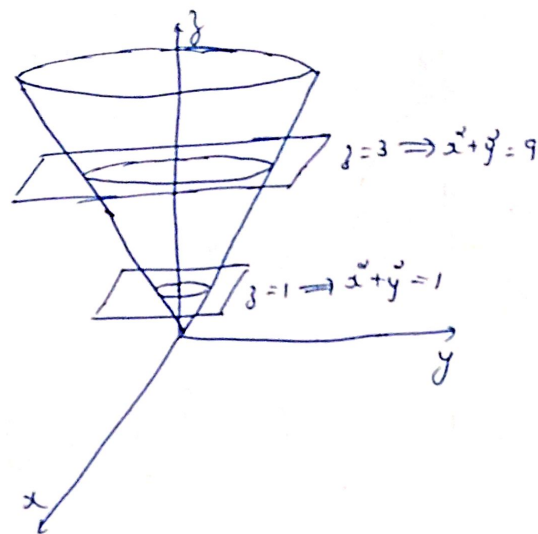
$$= \iint_R (x^2 + y^2 + 1) dA = \int_{\theta=0}^{2\pi} \int_{r=0}^1 (r^2 + 1) r \, dr \, d\theta$$

$$= \int_0^{2\pi} \left[\frac{r^4}{4} + \frac{r^2}{2} \right]_0^1 d\theta = \frac{3}{4} [0]_0^{2\pi} = \underline{\underline{\frac{3\pi}{2}}}$$

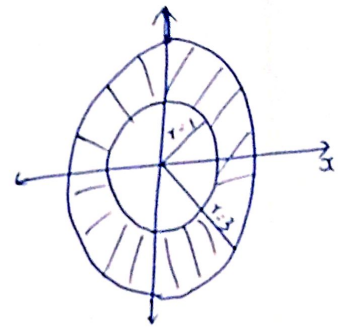
2) Let σ be the portion of the cone $z^2 = x^2 + y^2$ between the planes $z=1$ and $z=3$ oriented by upward unit normal. Find the flux of the vector field $\vec{F}(x, y, z) = x\hat{i} + y\hat{j} + z\hat{k}$.

Soln: $\sigma: z^2 = x^2 + y^2 \implies \frac{\partial z}{\partial x} = \frac{x}{\sqrt{x^2 + y^2}} \text{ and } \frac{\partial z}{\partial y} = \frac{y}{\sqrt{x^2 + y^2}}$

(12)



Projection R



$$\text{Flux } \phi = \iint_{\sigma} \vec{F} \cdot \hat{n} \, dS = \iint_R ((x\hat{i} + y\hat{j} + z\hat{k}) \cdot (-\hat{i} \frac{x}{\sqrt{x^2+y^2}} - \hat{j} \frac{y}{\sqrt{x^2+y^2}} + \hat{k})) \, dA$$

$$= \iint_R \left(\frac{-x^2}{\sqrt{x^2+y^2}} - \frac{y^2}{\sqrt{x^2+y^2}} + z \right) \, dA$$

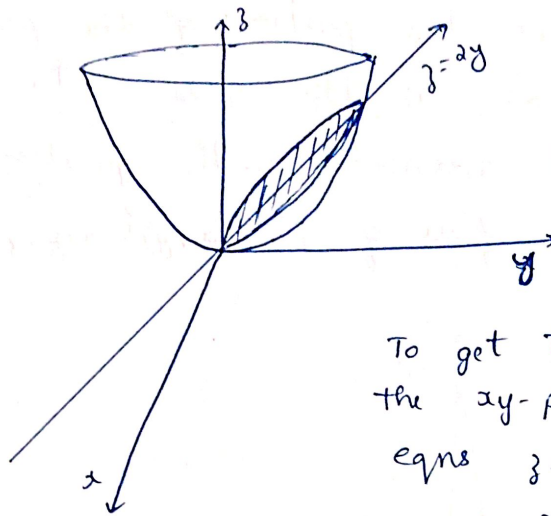
$$= \iint_R \left(\frac{-x^2-y^2}{\sqrt{x^2+y^2}} + z \sqrt{x^2+y^2} \right) \, dA$$

$$= \iint_R \left(\frac{x^2+y^2}{\sqrt{x^2+y^2}} \right) \, dA = \int_{\theta=0}^{2\pi} \int_{r=1}^3 \left(\frac{r^2}{r} \right) r \, dr \, d\theta$$

$$= \int_0^{2\pi} \left[\frac{r^3}{3} \right]_1^3 \, d\theta = \frac{26}{3} [0]_0^{2\pi} = \underline{\underline{\frac{52\pi}{3}}}$$

- 3) Let σ be the portion of the paraboloid $z = x^2 + y^2$ below the plane $z = 2y$ oriented by downward unit normal. Find the flux of the vector field $\vec{F} = x\hat{k}$

(13)

Soln:

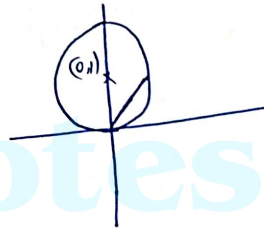
To get the projection R on the xy -plane, solve the eqns $z = x^2 + y^2$ and $z = 2y$

$$\Rightarrow 2y = x^2 + y^2$$

$$\Rightarrow (x-0)^2 + (y-1)^2 = 1$$

Circle $\rightarrow$ centre $(0,1)$

Radius = 1



in polar form,

eqn is

$$x^2 + y^2 - 2y = 0$$

$$r^2 - 2r \sin \theta = 0$$

$$r(r - 2 \sin \theta) = 0$$

$$\Rightarrow r = 0, r = 2 \sin \theta$$

$$\therefore \text{Flux } \phi = \iint_{\sigma} \vec{F} \cdot \hat{n} \, dS = \iint_R x \hat{k} \cdot \left(\hat{i} \frac{\partial z}{\partial x} + \hat{j} \frac{\partial z}{\partial y} - \hat{k} \right) dA$$

$$= \iint_R x \hat{k} \cdot (\hat{i}(2x) + \hat{j}(2y) - \hat{k}) dA$$

$$= \iint_R (-x) dA = \int_{\theta=0}^{\pi} \int_{r=0}^{2 \sin \theta} (-r \cos \theta) r \, dr \, d\theta$$

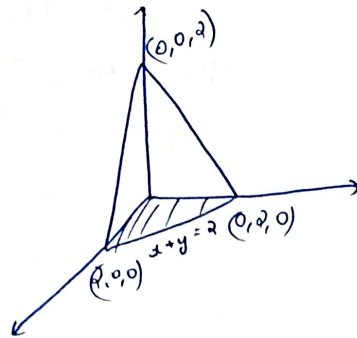
$$= \int_0^{\pi} -\cos \theta \left(\frac{r^3}{3} \right)_0^{2 \sin \theta} d\theta = \int_0^{\pi} -\cos \theta \left(\frac{8 \sin^3 \theta}{3} \right) d\theta$$

$$= -\frac{8}{3} \int_0^{\pi} u^3 du = \underline{\underline{0}}$$

Put $\sin \theta = u$
 $\cos \theta d\theta = du$

(14)

- 4) Let σ be the portion of the plane $x+y+z=2$ in the first octant, oriented by unit normals with positive side. Find the flux of $\vec{F} = (x+y)\hat{i} + (y+z)\hat{j} + (z+x)\hat{k}$

Soln :

$$\sigma: z = 2-x-y$$

$$\frac{\partial z}{\partial x} = -1$$

$$\frac{\partial z}{\partial y} = -1$$

$$\phi = \iint_{\sigma} \vec{F} \cdot \hat{n} \, dS = \iint_R [(x+y)\hat{i} + (y+z)\hat{j} + (z+x)\hat{k}] \cdot [-\hat{i}(-1) - \hat{j}(-1) + \hat{k}] \, dA$$

$$= \iint_R [(x+y) + (y+z) + (z+x)] \, dA$$

$$= \iint_R [x+y + y + (2-x-y) + (2-x-y) + x] \, dA$$

$$= \iint_R 4 \, dA = \int_{x=0}^2 \int_{y=0}^{2-x} 4 \, dy \, dx = 4 \int_0^2 [y]_0^{2-x} \, dx$$

$$= 4 \int_0^2 (2-x) \, dx$$

$$= 4 \left[2x - \frac{x^2}{2} \right]_0^2 = \underline{\underline{8}}$$

H.W

Ex 15.6, Question Nos: 7, 10, 12, 23, 24

Exercise 15.6

Find the flux of the vector field $\vec{F}$ across σ .

7) $\vec{F}(x, y, z) = x\hat{i} + y\hat{j} + z\hat{k}$, σ is the portion of the surface $z = 1 - x^2 - y^2$ above the xy -plane oriented by upward normals.

10) $\vec{F}(x, y, z) = y\hat{j} + \hat{k}$, σ is the portion of the paraboloid $z = x^2 + y^2$ below the plane $z = 4$, oriented by downward unit normals.

12) $\vec{F}(x, y, z) = x^2\hat{i} + yx\hat{j} + zx\hat{k}$, σ is the portion of the plane $6x + 3y + z = 6$ in the first octant, oriented by unit normals with positive components.

23) $\vec{F}(x, y, z) = \hat{i} + \hat{j} + \hat{k}$, σ is the portion of the cone $z = \sqrt{x^2 + y^2}$ between the planes $z = 1$ and $z = 2$ oriented by downward unit normals.

24) $\vec{F}(x, y, z) = x\hat{i} + y\hat{j} + z\hat{k}$, σ is the portion of the cylinder $x^2 + z^2 = 1$ between the planes $y = 1$ and $y = -2$, oriented by outward unit normals.

(15)

Gauss Divergence Theorem

Let V be a solid whose surface σ is oriented outward. If

$$F(x, y, z) = f(x, y, z)\hat{i} + g(x, y, z)\hat{j} + h(x, y, z)\hat{k}$$

where f, g, h have continuous first order partial derivatives on some open set containing V and if n is the outward unit normal on σ ,

$$\text{then, } \iint_{\sigma} \vec{F} \cdot \hat{n} \, dS = \iiint_V (\text{div } F) \, dV = \iiint_V \left(\frac{\partial f}{\partial x} + \frac{\partial g}{\partial y} + \frac{\partial h}{\partial z} \right) dV$$

1. Use divergence theorem to find the outward flux of the vector field $F(x, y, z) = 2x\hat{i} + 3y\hat{j} + z\hat{k}$ across the unit cube.

Soln: Flux $\phi = \iint_{\sigma} \vec{F} \cdot \hat{n} \, dS = \iiint_V (\text{div } F) \, dV$

$$= \iiint_V \left[\frac{\partial}{\partial x}(2x) + \frac{\partial}{\partial y}(3y) + \frac{\partial}{\partial z}(z) \right] dV$$

$$= \int_{z=0}^1 \int_{y=0}^1 \int_{x=0}^1 (5 + z) \, dx \, dy \, dz$$

$$= \int_0^1 \int_0^1 (5 + z) [x]_0^1 \, dy \, dz$$

$$= \int_0^1 (5 + z) [y]_0^1 \, dz = [5z + \frac{z^2}{2}]_0^1$$

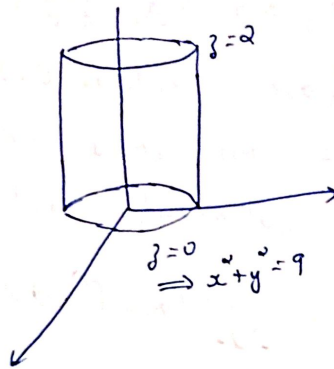
$$= \underline{\underline{6}}$$

- 2) Use divergence theorem to find the outward flux of the vector field $F(x, y, z) = x^3\hat{i} + y^3\hat{j} + z^3\hat{k}$

(16)

across the surface of the region that is enclosed by the circular cylinder $x^2 + y^2 = 9$ and the planes $z = 0$ and $z = 2$.

Soln :-



$$\begin{aligned}
 \phi &= \iiint_V (\text{div } \vec{F}) dV = \iiint_V (3x^2 + 3y^2 + 2z) dV \\
 &= \int_{\theta=0}^{2\pi} \int_{r=0}^3 \int_{z=0}^2 (3r^2 + 2z) r dz dr d\theta \quad (\text{Using cylindrical co-ordinates}) \\
 &= \int_0^{2\pi} \int_0^3 \left(3r^3 z + z^2 r \right) \Big|_0^2 dr d\theta \\
 &= \int_0^{2\pi} \int_0^3 (6r^3 + 4r) dr d\theta = \int_0^{2\pi} \left(\frac{6r^4}{4} + \frac{4r^2}{2} \right) \Big|_0^3 d\theta \\
 &= \frac{279}{2} [0]_0^{2\pi} = \underline{\underline{279\pi}}
 \end{aligned}$$

3) Use divergence theorem to find the outward flux of the vector field $\vec{F}(x, y, z) = z\hat{k}$ across the sphere $x^2 + y^2 + z^2 = a^2$.

Soln :- $\phi = \iiint_V (\text{div } \vec{F}) dV = \iiint_V dV = \text{Volume of the sphere}$

$$= \underline{\underline{\frac{4}{3}\pi a^3}}$$

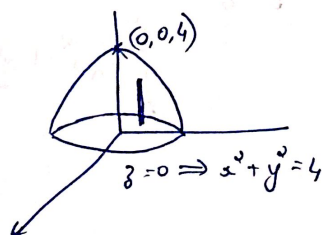
(17)

- 4) Use divergence theorem to find the outward flux of the vector field $F(x, y, z) = x^3 \hat{i} + y^3 \hat{j} + z^3 \hat{k}$ across the surface of the region that is enclosed by the hemisphere $z = \sqrt{a^2 - x^2 - y^2}$ and the plane $z = 0$.

$$\begin{aligned}
 \text{Soln} \div \phi &= \iiint_V (\text{div } F) dV = \iiint_V (3x^2 + 3y^2 + 3z^2) dV \\
 &= 3 \int_{\theta=0}^{2\pi} \int_{\phi=0}^{\pi/2} \int_{\rho=0}^a (\rho^2) \rho^2 \sin \phi \, d\rho \, d\phi \, d\theta \quad (\text{Using spherical coordinates}) \\
 &= 3 \int_0^{2\pi} \int_0^{\pi/2} \sin \phi \left[\frac{\rho^5}{5} \right]_0^a d\phi \, d\theta \\
 &= 3 \int_0^{2\pi} \int_0^{\pi/2} \left(\frac{a^5}{5} \sin \phi \right) d\phi \, d\theta = \frac{3a^5}{5} \int_0^{2\pi} (-\cos \phi)_0^{\pi/2} d\theta \\
 &= \frac{3a^5}{5} [0]_0^{2\pi} = \underline{\underline{\frac{6\pi a^5}{5}}}
 \end{aligned}$$

- 5) Use DT to find the outward flux of the vector field $F(x, y, z) = x \hat{i} + y \hat{j} + z \hat{k}$ across the surface of the solid bounded by the paraboloid $z = 4 - x^2 - y^2$ and the xy -plane.

Soln :

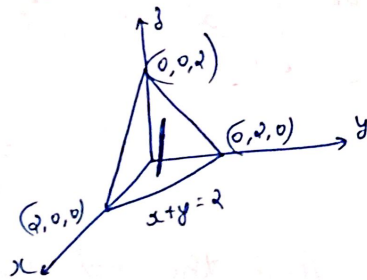


(18)

$$\begin{aligned}
 \phi &= \iiint_V (\operatorname{div} F) dV = \iiint_V 3 dV = 3 \int_0^{2\pi} \int_0^2 \int_0^{4-x^2-y^2} r dz dr d\theta \\
 &= 3 \int_0^{2\pi} \int_0^2 r [z]_0^{4-x^2-y^2} dr d\theta \\
 &= 3 \int_0^{2\pi} \int_0^2 (4r - r^3) dr d\theta = 3 \int_0^{2\pi} \left[2r^2 - \frac{r^4}{4} \right]_0^2 d\theta \\
 &= 3 \int_0^{2\pi} 4 d\theta = 12 [0]_0^{2\pi} = \underline{\underline{24\pi}}
 \end{aligned}$$

- 6) Use DT to find the outward flux of the vector field $\vec{F}(x, y, z) = (x^2 + y)\hat{i} + xy\hat{j} - (xz + y)\hat{k}$ across the surface of the tetrahedron in the first octant bounded by $x + y + z = 2$ and the co-ordinate planes.

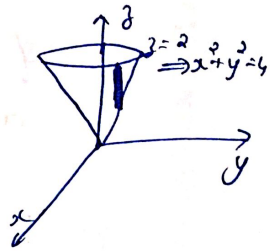
Soln :



$$\begin{aligned}
 \phi &= \iiint_V (\operatorname{div} F) dV = \iiint_V x dV = \int_0^2 \int_0^{2-x} \int_0^{2-x-y} x dz dy dx \\
 &= \int_0^2 \int_0^{2-x} (2x - x^2 - xy) dy dx = \int_0^2 \left(2xy - \frac{x^2 y}{2} - x \frac{y^2}{2} \right) \Big|_0^{2-x} dx \\
 &= \int_0^2 \left(-2x^2 + \frac{x^3}{2} + 2x \right) dx = \underline{\underline{\frac{2}{3}}}
 \end{aligned}$$

(19)

→ Use DT to find the outward flux of the vector field $\vec{F}(x, y, z) = x\hat{i} + y\hat{j} + z\hat{k}$ across the surface of the conical solid bounded by $z = \sqrt{x^2 + y^2}$ and $z = 2$



Soln : $\phi = \iiint_V (\text{div } \vec{F}) dv$

$$= 2 \iiint_V (x + y + z) dv$$

$$= 2 \int_0^{2\pi} \int_0^2 \int_{z=\sqrt{x^2+y^2}}^2 (r \cos \theta + r \sin \theta + z) r dz dr d\theta$$

$$= 2 \int_0^{2\pi} \int_0^2 \left[r^2 (\cos \theta + \sin \theta) z + \frac{r^3}{3} \right]_r^2 d\theta$$

$$= 2 \int_0^{2\pi} \left[\frac{2r^3}{3} (\cos \theta + \sin \theta) + r^3 - \frac{r^4}{4} (\cos \theta + \sin \theta) - \frac{r^4}{8} \right]_0^2 d\theta$$

$$= 2 \int_0^{2\pi} \left[(\cos \theta + \sin \theta) \left(\frac{4}{3} \right) + 2 \right] d\theta$$

$$= 2 \left[\frac{4}{3} (\sin \theta - \cos \theta) + 2\theta \right]_0^{2\pi} = \underline{\underline{8\pi}}$$

H.w

Ex 15.7 Question Nos: 3, 4, 11, 13, 18, 19

(20)

H.W.

Use Divergence Theorem to find the outward flux of the vector field

1) $\vec{F}(x, y, z) = x\hat{i} + y\hat{j} + z\hat{k}$; σ is the surface of the cube bounded by the planes $x=0$, $x=1$, $y=0$, $y=1$, $z=0$ and $z=1$

2) $\vec{F}(x, y, z) = 4x\hat{i} - 2y^2\hat{j} + z^2\hat{k}$ taken over the region bounded by the cylinder $x^2 + y^2 = 4$ and the planes $z=0$ and $z=3$.

3) $\vec{F}(x, y, z) = (x^2 + y)\hat{i} + z^2\hat{j} + (e^y - z)\hat{k}$ where σ is the surface of the rectangular solid bounded by the co-ordinate planes and the planes $x=3$, $y=1$ and $z=3$.

Exercise 15.7

- 3) Use divergence theorem to find the outward flux of the vector field

$\vec{F}(x, y, z) = 2x\hat{i} - yz\hat{j} + z^2\hat{k}$, the surface σ is the paraboloid $z = x^2 + y^2$ capped by the disk $x^2 + y^2 \leq 1$ in the plane $z = 1$

- 4) Use divergence theorem to find the outward flux of the vector field

$\vec{F}(x, y, z) = xy\hat{i} + yz\hat{j} + xz\hat{k}$, σ is the surface of the cube bounded by the planes $x=0, x=2, y=0, y=2, z=0, z=2$

- 11) Use divergence theorem to find the outward flux of the vector field

$\vec{F}(x, y, z) = (x-z)\hat{i} + (y-x)\hat{j} + (z-y)\hat{k}$; σ is the surface of the cylindrical solid bounded by $x^2 + y^2 = a^2$, $z=0$ and $z=1$.

- 13) Use divergence theorem to find the outward flux of the vector field $\vec{F}(x, y, z) = x^3\hat{i} + y^3\hat{j} + z^3\hat{k}$, σ is the surface of the cylindrical solid $x^2 + y^2 = 4$, $z=0$ and $z=3$

- 18) Use divergence theorem to find the outward flux of the vector field $\vec{F}(x, y, z) = x^2y\hat{i} - xy^2\hat{j} + (z+2)\hat{k}$, σ is the

surface of the solid bounded above by the plane $z = 2x$ and below by the paraboloid $z = x^2 + y^2$.

- 19) Use divergence theorem to find the outward flux of the vector field $\vec{F}(x, y, z) = x^3 \hat{i} + xy \hat{j} + yz \hat{k}$; σ is the surface of the solid bounded by $z = 4 - x^2$, $y + z = 5$, $z = 0$ and $y = 0$.

Ktunotes.in

(21)

Stoke's theorem

Let σ be a piecewise smooth oriented surface that is bounded by a simple, closed, piecewise smooth curve C . If the components of the vector field

$$\vec{F}(x, y, z) = f(x, y, z)\hat{i} + g(x, y, z)\hat{j} + h(x, y, z)\hat{k}$$

are continuous and have continuous first order partial derivatives on some open set containing σ , then

$$\oint_C \vec{F} \cdot d\vec{r} = \iint_{\sigma} (\text{curl } \vec{F}) \cdot \hat{n} \, dS$$

where n is the unit normal towards the positive side of σ .

1. Use Stoke's theorem to evaluate $\oint_C \vec{F} \cdot d\vec{r}$

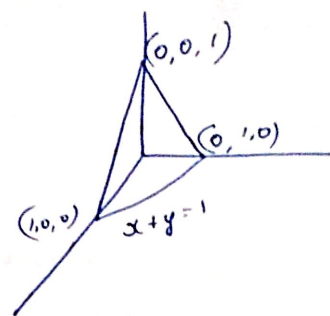
where $\vec{F}(x, y, z) = (x-y)\hat{i} + (y-z)\hat{j} + (z-x)\hat{k}$ where

C is the boundary of the portion of the plane $x+y+z=1$ in the first octant. Assume that the surface has an upward direction.

$$\text{Soln:- } \text{curl } \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ (x-y) & (y-z) & (z-x) \end{vmatrix} = \hat{i} + \hat{j} + \hat{k}$$

$$\begin{aligned} \oint_C \vec{F} \cdot d\vec{r} &= \iint_{\sigma} \text{curl } \vec{F} \cdot \hat{n} \, dS = \iint_R (\hat{i} + \hat{j} + \hat{k}) \cdot \left(-\hat{i} \frac{\partial z}{\partial x} - \hat{j} \frac{\partial z}{\partial y} + \hat{k} \right) dA \\ &= \iint_R (\hat{i} + \hat{j} + \hat{k}) \cdot (\hat{i} + \hat{j} + \hat{k}) dA \quad \left| \begin{array}{l} z = 1-x-y \\ \frac{\partial z}{\partial x} = -1 \\ \frac{\partial z}{\partial y} = -1 \end{array} \right. \end{aligned}$$

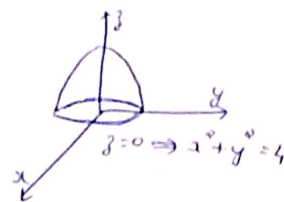
$$\begin{aligned}
 & \text{(22)} \\
 &= 3 \int_{x=0}^1 \int_{y=0}^{1-x} dy dx \\
 &= 3 \int_0^1 [y]_0^{1-x} dx = 3 \int_0^1 (1-x) dx \\
 &= 3 \left[x - \frac{x^2}{2} \right]_0^1 = \underline{\underline{\frac{3}{2}}}
 \end{aligned}$$



2) Use Stoke's Theorem to evaluate $\oint_C \vec{F} \cdot d\vec{r}$ where $F(x,y,z) = 2z\hat{i} + 3x\hat{j} + 5y\hat{k}$ and σ to be the portion of the paraboloid $z = 4 - x^2 - y^2$ for which $z \geq 0$ with upward direction and C to be positively oriented circle $x^2 + y^2 = 4$ that forms the boundary of σ in the xy -plane.

Soln:-

$$\begin{aligned}
 \oint_C \vec{F} \cdot d\vec{r} &= \iint_{\sigma} \text{curl } \vec{F} \cdot \hat{n} \, dS \\
 &= \iint_R \text{curl } \vec{F} \cdot \left(-\hat{i} \frac{\partial z}{\partial x} - \hat{j} \frac{\partial z}{\partial y} + \hat{k} \right) dA
 \end{aligned}$$



$$\text{curl } \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 2z & 3x & 5y \end{vmatrix} = 5\hat{i} + 2\hat{j} + 3\hat{k}$$

$$z = 4 - x^2 - y^2 \implies \frac{\partial z}{\partial x} = -2x \quad \text{and} \quad \frac{\partial z}{\partial y} = -2y$$

(23)

$$\begin{aligned}
\oint_C \vec{F} \cdot d\vec{r} &= \iint_R (5\hat{i} + 2\hat{j} + 3\hat{k}) \cdot (2x\hat{i} + 2y\hat{j} + \hat{k}) dA \\
&= \iint_R (10x + 4y + 3) dA = \int_{\theta=0}^{2\pi} \int_{r=0}^2 (10r\cos\theta + 4r\sin\theta + 3) r dr d\theta \\
&= \int_0^{2\pi} \left[\frac{10r^3}{3} \cos\theta + \frac{4r^3}{3} \sin\theta + \frac{3r^2}{2} \right]_0^2 d\theta \\
&= \int_0^{2\pi} \left(\frac{80}{3} \cos\theta + \frac{32}{3} \sin\theta + 6 \right) d\theta \\
&= \left[\frac{80}{3} \sin\theta - \frac{32}{3} \cos\theta + 6\theta \right]_0^{2\pi} = \underline{\underline{12\pi}}
\end{aligned}$$

3) Use Stokes's theorem to evaluate $\oint_C \vec{F} \cdot d\vec{r}$
 where $\vec{F}(x, y, z) = z^2\hat{i} + 3xz\hat{j} - y^3\hat{k}$ and C is the
 circle $x^2 + y^2 = 1$ in the xy -plane with counter
 clockwise direction looking down the $+ve$ z -axis.

Soln : $\text{Curl } \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ z^2 & 3xz & -y^3 \end{vmatrix} = (-3y^2)\hat{i} + (2z)\hat{j} + 3\hat{k}$

$$z=0 \Rightarrow \frac{\partial z}{\partial x} = 0 \quad \text{and} \quad \frac{\partial z}{\partial y} = 0.$$

$$\begin{aligned}
\therefore \oint_C \vec{F} \cdot d\vec{r} &= \iint_R \text{Curl } \vec{F} \cdot \left(-\hat{i} \frac{\partial z}{\partial x} - \hat{j} \frac{\partial z}{\partial y} + \hat{k} \right) dA \\
&= \iint_R [(-3y^2)\hat{i} + (2z)\hat{j} + 3\hat{k}] \cdot \hat{k} dA \\
&= \iint_R 3 dA = 3 \int_{\theta=0}^{2\pi} \int_{r=0}^1 r dr d\theta = 3 \int_0^{2\pi} \frac{1}{2} d\theta = \underline{\underline{3\pi}}
\end{aligned}$$

(24)

- 4) Use Stokes' theorem to evaluate $\oint_C \vec{F} \cdot d\vec{r}$ where $\vec{F}(x, y, z) = -3y^2 \hat{i} + 4z \hat{j} + 6x \hat{k}$ and C is the triangle in the plane $z = \frac{y}{2}$ with vertices $(0, 0, 0)$, $(0, 2, 1)$ and $(2, 0, 0)$ with a counter clockwise direction looking down the +ve z -axis.

Soln:- $\text{curl } \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ -3y^2 & 4z & 6x \end{vmatrix} = -4\hat{i} - 6\hat{j} + 6y\hat{k}$

$$z = \frac{y}{2} \Rightarrow \frac{\partial z}{\partial x} = 0 \text{ and } \frac{\partial z}{\partial y} = \frac{1}{2}$$

$$\oint_C \vec{F} \cdot d\vec{r} = \iint_R (-4\hat{i} - 6\hat{j} + 6y\hat{k}) \cdot \left(\frac{-1}{2}\hat{j} + \hat{k}\right) dA$$

$$= \iint_R (3 + 6y) dA$$

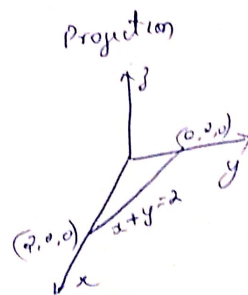
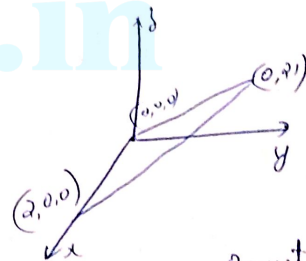
$$= \int_{x=0}^2 \int_{y=0}^{2-x} (3 + 6y) dy dx$$

$$= \int_0^2 (3y + 3y^2)_0^{2-x} dx$$

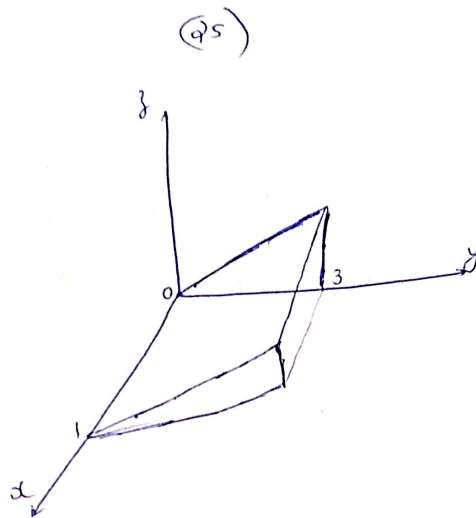
$$= \int_0^2 (3x^2 - 15x + 18) dx = \left[x^3 - \frac{15x^2}{2} + 18x \right]_0^2$$

$$= \underline{\underline{14}}$$

- 5) Use Stokes' theorem to evaluate $\oint_C \vec{F} \cdot d\vec{r}$ where $\vec{F}(x, y, z) = x^2 \hat{i} + 4xy^3 \hat{j} + y^2 x \hat{k}$ and C is the rectangle $0 \leq x \leq 1$, $0 \leq y \leq 3$ in the plane $z = y$ in the upward direction.



Soln:



$$\text{Curl } \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x^2 & 4xy^3 & y^2x \end{vmatrix} = (2xy)\hat{i} - (y^2)\hat{j} + (4y^3)\hat{k}$$

$$z = y \Rightarrow \frac{\partial z}{\partial x} = 0 \text{ and } \frac{\partial z}{\partial y} = 1$$

$$\begin{aligned} \oint_C \vec{F} \cdot d\vec{r} &= \iint_R [(2xy)\hat{i} - (y^2)\hat{j} + (4y^3)\hat{k}] \cdot (-\hat{j} + \hat{k}) dA \\ &= \iint_R (y^2 + 4y^3) dA = \int_{y=0}^3 \int_{x=0}^1 (y^2 + 4y^3) dx dy \\ &= \int_0^3 (y^2 + 4y^3) (x)_0^1 dy \\ &= \left[\frac{y^3}{3} + y^4 \right]_0^3 = \underline{\underline{90}} \end{aligned}$$

H.W Ex 15.8

I) Use Stoke's theorem to evaluate $\oint_C \vec{F} \cdot d\vec{r}$ where
 i) $\vec{F} = x^2\hat{i} + y^2\hat{j} + z^2\hat{k}$; σ is the portion of
 the cone $z = \sqrt{x^2 + y^2}$ below the plane $z=1$

(26)

a) $\vec{F} = (z-y)\hat{i} + (z+x)\hat{j} - (x+y)\hat{k}$;

σ is the portion of the paraboloid

$z = 4 - x^2 - y^2$ above the xy -plane.

b) $\vec{F} = x\hat{i} + y\hat{j} + z\hat{k}$; σ is the

upper hemisphere $z = \sqrt{a^2 - x^2 - y^2}$.

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